# Lecture 11: Describing Relationships

## Problem Set for Lecture 11

These questions will assist in bolstering your understanding of the material in Lecture 11. Emphasis should be placed on having the correct code/output as well as communication. The deliverable should be a knitted RMarkdown document to pdf without any code running off the page. You will be able to present these in office hours as an oral assessment interview along with the problem set for Lecture 12 and Lecture 13.

***** First, make sure you have read the Lecture Material's writeup *****

1. Using the `Batting` and `People` datasets in the `Lahman` package, create a new dataset called `baseball` by joining the two datasets together and keeping the variables needed for this problem. Then restrict your data to seasons from 1920 or later and to players with at least 100 at-bats in a season. Use this dataset to investigate how offensive performance is related to age, decade, and birthplace. You can remove any missing values that are present. Then answer the following questions.

   (a) Determine the players with the largest total home run counts in this filtered dataset. Display the player ID, first name, last name, and total number of home runs, and identify the top 10.

   (b) For each age, determine the total number of observations and the overall batting average, where batting average is defined by total hits (H) divided by total at-bats (AB). Restrict your attention to ages with at least 10 observations.

   (c) Using the summarized data from the previous part, you might be interested in investigating the relationship between age and batting average. You calculate the correlation as -0.0114 and notice that the visualization shows a non-linear relationship. Because of that you decide to compare the relationship for players less than 30 and those 30 years or older by creating the visualization (found below).

   (d) Create an `age_group` variable with categories `under_25`, `25_29`, `30_34`, as well as `35_or_older`. Also create a `decade` variable from `yearID`. Then create the visualization (found below) that compares the distribution of batting averages across decades for the different age groups.

   (e) Create a summary table showing the mean batting average for each age-group within each decade, where each decade is a row and each age group is its own column.

   (f) Determine the five most common birth states in the dataset. Then, using only those five states, investigate whether birth state and decade appear to be associated. Use counts and conditional proportions to recreate the visualization (found below) to support your conclusion.

2. Using the `Birthdays` dataset in the `mosaicData` package, investigate how birth counts vary across the calendar and across states. Then answer the following questions.

   (a) Determine the average number of births for each day of the week. Which day has the largest average number of births and which has the smallest? (Sunday is the least, Tuesday is the most)

   (b) Compute the mean number of births for each year-month-day-of-week combination. Create the visualization (found below) that compares these mean birth counts across the days of the week.

   (c) Create a table showing the total number of births for each combination of month and day of the week.

   (d) Determine the percentage of all births that occur on each month-day combination. Then create the visualization (found below) that displays these percentages across the calendar. Within the `scale_fill_steps2()` function, use the following layer within the ggplot plot creation:

```
scale_fill_steps2(low = "red",
                  mid = "white",
                  high = "green",
                  midpoint = 0.275,
                  limits = c(0.22, 0.30),
                  oob = scales::squish,
                  breaks = seq(0.22, 0.30, by = 0.01))
```

   (e) For each state, determine the total number of births in 1970 and in 1988. Use this information to study the relationship between births in 1970 and births in 1988 across states. Create the visualization (found below), calculate the correlation, and describe the relationship in terms of direction, form, strength, and unusual features.

   (f) Using only the locations `DC`, `PA`, `MD`, and `VA`, determine, for each year, the proportion of births contributed by each location out of the total births among those four locations. Then create the visualization (found below) that shows how these proportions change over time.

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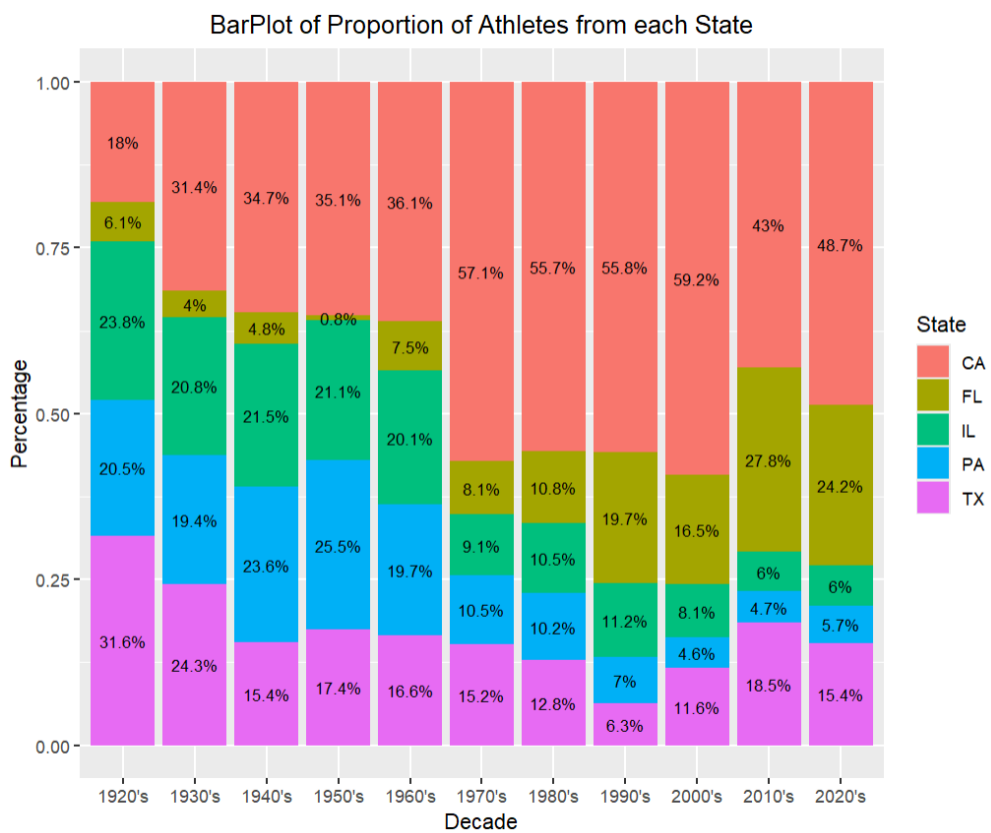
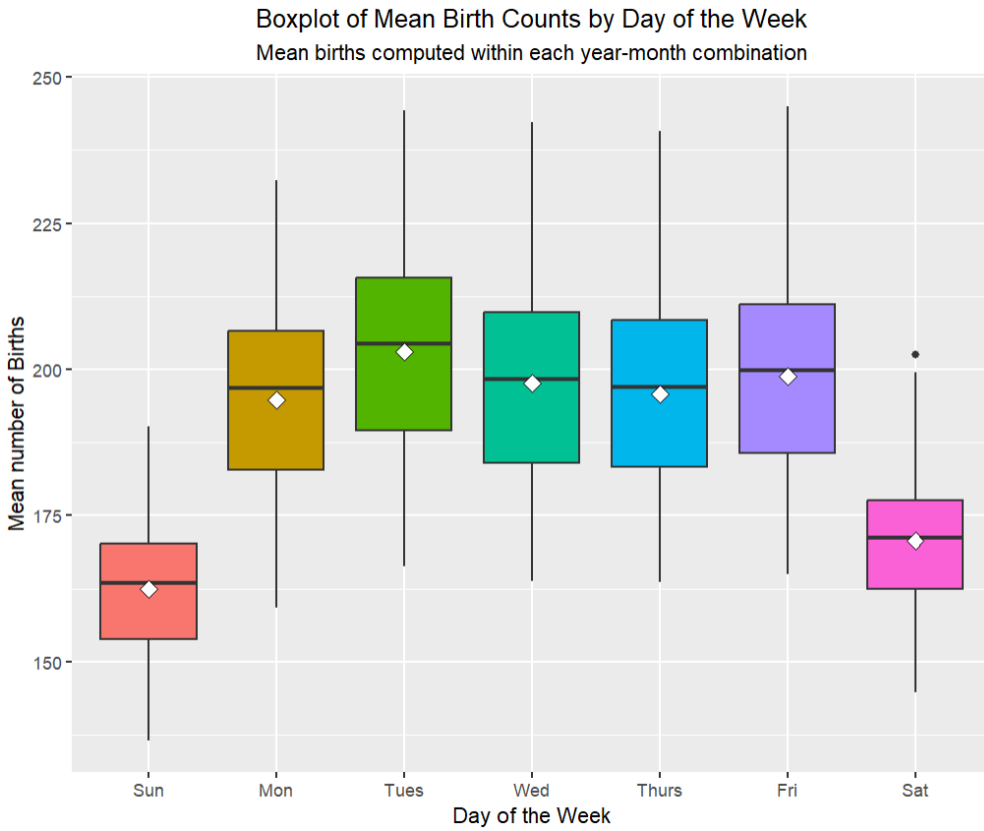
**Visualizations and Tables below to Assist you**

Correlation between Age and Batting AVG
Based on Age Classification (cor= 0.534 and -0.666)
AVG
0.270
0.265
0.260
20
25
30
35
40
45
Age

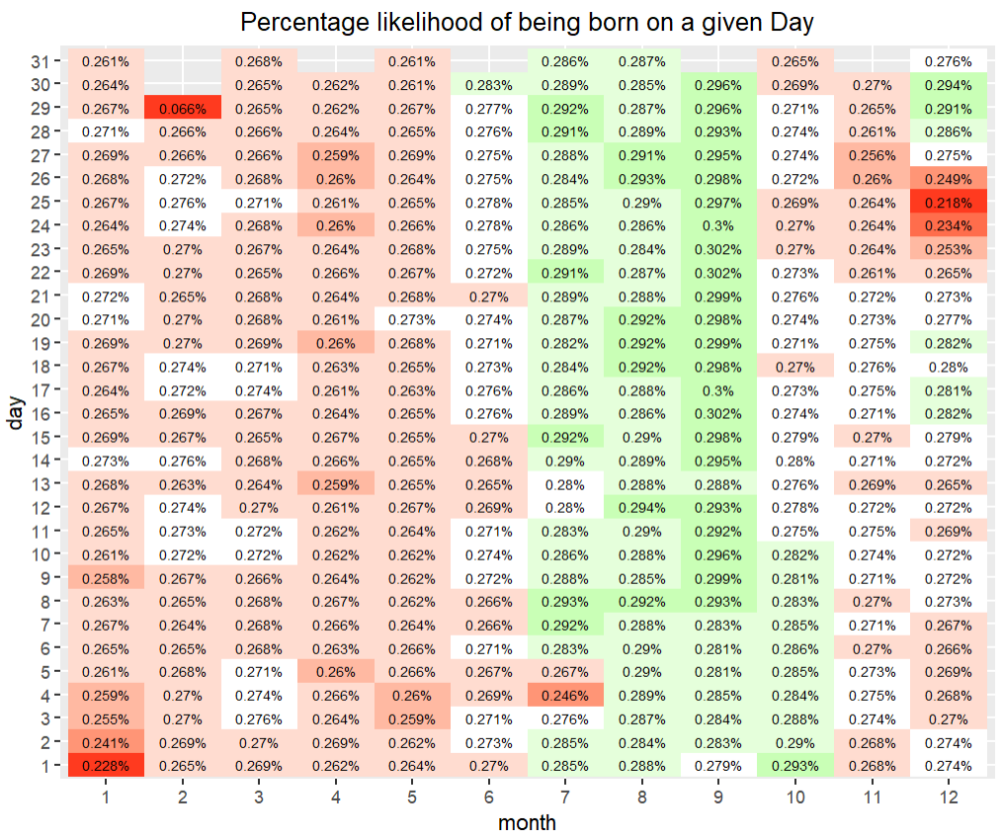
Boxplot of Batting AVG for each Decade
Based on the Age Group
Age Group
under_25
25_29
30_34
35_or_older
AVG
0.4
0.3
0.2
0.1
1920's
1930's
1940's
1950's
1960's
1970's
1980's
1990's
2000's
2010's
2020's
Decade

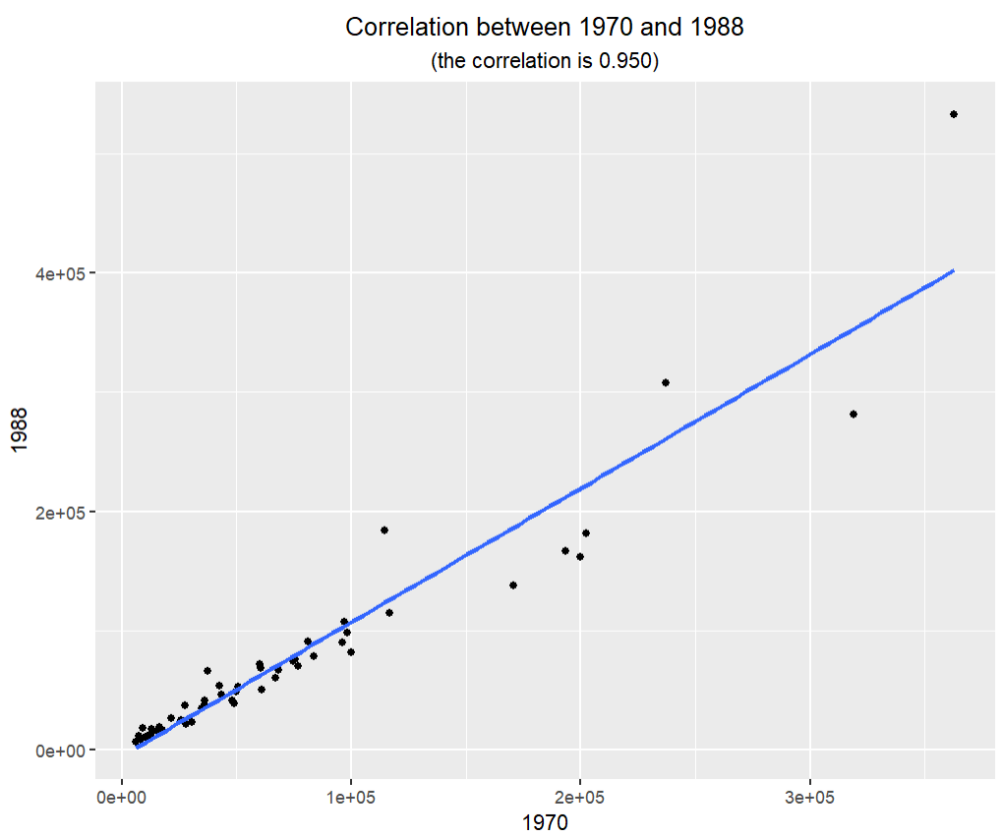
```
   decade under_25 `25_29` `30_34` `35_or_older`
   <chr>     <dbl>   <dbl>   <dbl>         <dbl>
 1 1920's    0.283   0.287   0.285         0.287
 2 1930's    0.284   0.282   0.284         0.284
 3 1940's    0.266   0.266   0.266         0.272
 4 1950's    0.257   0.262   0.263         0.265
 5 1960's    0.249   0.251   0.250         0.255
 6 1970's    0.253   0.256   0.256         0.257
 7 1980's    0.255   0.258   0.262         0.255
 8 1990's    0.259   0.263   0.265         0.268
 9 2000's    0.265   0.265   0.265         0.264
10 2010's    0.256   0.253   0.253         0.251
11 2020's    0.242   0.242   0.244         0.241
```

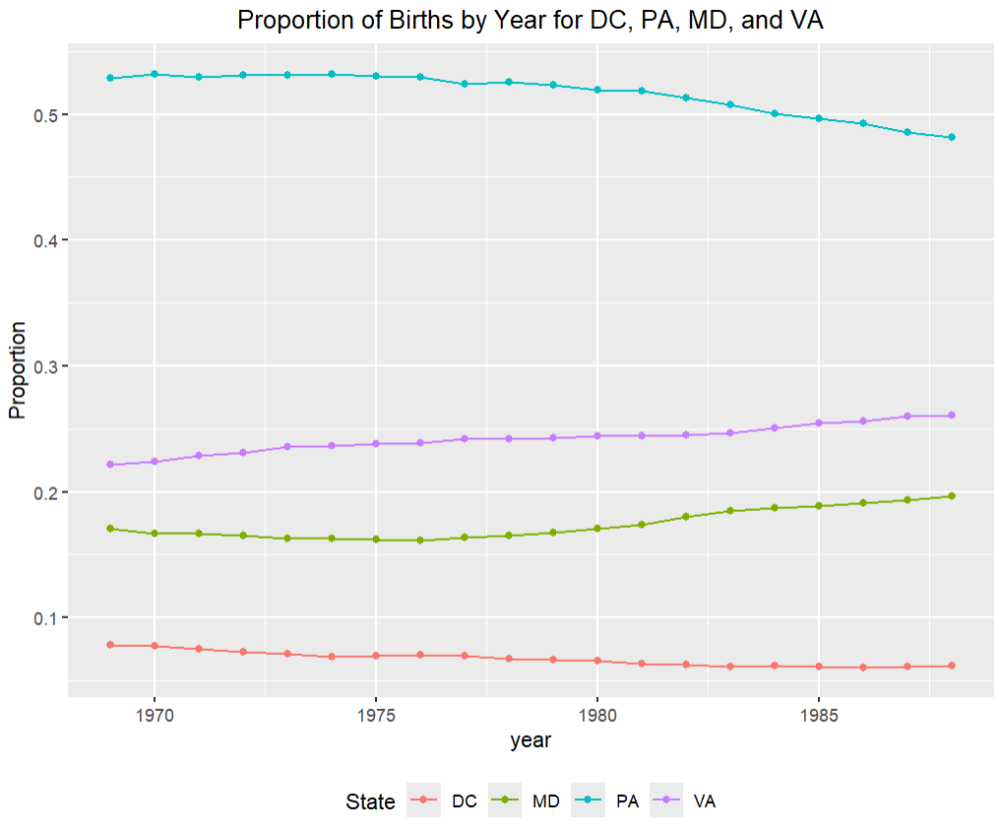





```
   month    Sun    Mon   Tues    Wed  Thurs    Fri    Sat
   <int>  <int>  <int>  <int>  <int>  <int>  <int>  <int>
 1     1 714590 836139 857521 845615 861021 878440 765839
 2     2 666610 791305 818203 799951 786359 807704 692814
 3     3 723811 883247 904828 867897 863051 864440 760866
 4     4 665550 815348 860751 843635 830436 836368 708815
 5     5 711177 830672 885864 851412 866788 882860 756447
 6     6 703844 864812 884425 864275 850923 858614 731803
 7     7 753943 889500 960908 937586 944654 938384 795320
 8     8 786658 935981 961855 934698 917682 952396 820494
 9     9 758555 891032 961394 937372 930827 932728 794265
10    10 742663 885689 906782 905599 903144 927794 783701
11    11 715932 863377 888776 845034 802886 843598 735885
12    12 703817 884917 922621 900465 877195 869998 745363
```


Percentage likelihood of being born on a given Day
day
month
31 0.261% 0.268% 0.261% 0.286% 0.287% 0.265% 0.276%
30 0.264% 0.265% 0.262% 0.261% 0.283% 0.289% 0.285% 0.296% 0.269% 0.27% 0.294%
29 0.267% 0.066% 0.265% 0.262% 0.267% 0.277% 0.292% 0.287% 0.296% 0.271% 0.265% 0.291%
28 0.271% 0.266% 0.266% 0.264% 0.265% 0.276% 0.291% 0.289% 0.293% 0.274% 0.261% 0.286%
27 0.269% 0.266% 0.266% 0.259% 0.269% 0.275% 0.288% 0.291% 0.295% 0.274% 0.256% 0.275%
26 0.268% 0.272% 0.268% 0.26% 0.264% 0.275% 0.284% 0.293% 0.298% 0.272% 0.26% 0.249%
25 0.267% 0.276% 0.271% 0.261% 0.265% 0.278% 0.285% 0.29% 0.297% 0.269% 0.264% 0.218%
24 0.264% 0.274% 0.268% 0.26% 0.266% 0.278% 0.286% 0.286% 0.3% 0.27% 0.264% 0.234%
23 0.265% 0.27% 0.267% 0.264% 0.268% 0.275% 0.289% 0.284% 0.302% 0.27% 0.264% 0.253%
22 0.269% 0.27% 0.265% 0.266% 0.267% 0.272% 0.291% 0.287% 0.302% 0.273% 0.261% 0.265%
21 0.272% 0.265% 0.268% 0.264% 0.268% 0.27% 0.289% 0.288% 0.299% 0.276% 0.272% 0.273%
20 0.271% 0.27% 0.268% 0.261% 0.273% 0.274% 0.287% 0.292% 0.298% 0.274% 0.273% 0.277%
19 0.269% 0.27% 0.269% 0.26% 0.268% 0.271% 0.282% 0.292% 0.299% 0.271% 0.275% 0.282%
18 0.267% 0.274% 0.271% 0.263% 0.265% 0.273% 0.284% 0.292% 0.298% 0.27% 0.276% 0.28%
17 0.264% 0.272% 0.274% 0.261% 0.263% 0.276% 0.286% 0.288% 0.3% 0.273% 0.275% 0.281%
16 0.265% 0.269% 0.267% 0.264% 0.265% 0.276% 0.289% 0.286% 0.302% 0.274% 0.271% 0.282%
15 0.269% 0.267% 0.265% 0.267% 0.265% 0.27% 0.292% 0.29% 0.298% 0.279% 0.27% 0.279%
14 0.273% 0.276% 0.268% 0.266% 0.265% 0.268% 0.29% 0.289% 0.295% 0.28% 0.271% 0.272%
13 0.268% 0.263% 0.264% 0.259% 0.265% 0.265% 0.28% 0.288% 0.288% 0.276% 0.269% 0.265%
12 0.267% 0.274% 0.27% 0.261% 0.267% 0.269% 0.28% 0.294% 0.293% 0.278% 0.272% 0.272%
11 0.265% 0.273% 0.272% 0.262% 0.264% 0.271% 0.283% 0.29% 0.292% 0.275% 0.275% 0.269%
10 0.261% 0.272% 0.272% 0.262% 0.262% 0.274% 0.286% 0.288% 0.296% 0.282% 0.274% 0.272%
9 0.258% 0.267% 0.266% 0.264% 0.262% 0.272% 0.288% 0.285% 0.299% 0.281% 0.271% 0.272%
8 0.263% 0.265% 0.268% 0.267% 0.262% 0.266% 0.293% 0.292% 0.293% 0.283% 0.27% 0.273%
7 0.267% 0.264% 0.268% 0.266% 0.264% 0.266% 0.292% 0.288% 0.283% 0.285% 0.271% 0.267%
6 0.265% 0.265% 0.268% 0.263% 0.266% 0.271% 0.283% 0.29% 0.281% 0.286% 0.27% 0.266%
5 0.261% 0.268% 0.271% 0.26% 0.266% 0.267% 0.267% 0.29% 0.281% 0.285% 0.273% 0.269%
4 0.259% 0.27% 0.274% 0.266% 0.26% 0.269% 0.246% 0.289% 0.285% 0.284% 0.275% 0.268%
3 0.255% 0.27% 0.276% 0.264% 0.259% 0.271% 0.276% 0.287% 0.284% 0.288% 0.274% 0.27%
2 0.241% 0.269% 0.27% 0.269% 0.262% 0.273% 0.285% 0.284% 0.283% 0.29% 0.268% 0.274%
1 0.228% 0.265% 0.269% 0.262% 0.264% 0.27% 0.285% 0.288% 0.279% 0.293% 0.268% 0.274%
1 2 3 4 5 6 7 8 9 10 11 12


Correlation between 1970 and 1988
(the correlation is 0.950)
1988
1970
0e+00
2e+05
4e+05
0e+00
1e+05
2e+05
3e+05


Proportion of Births by Year for DC, PA, MD, and VA
Proportion
0.5
0.4
0.3
0.2
0.1
1970
1975
1980
1985
year
State
DC
MD
PA
VA